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# Gravitational-Susceptibility Dip Behind DESI's results: Addressing the Phantom Crossing by Local limit of Nonlocal Gravity

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The second data release of the Dark Energy Spectroscopic Instrument (DESI), based on more than fourteen million galaxies and quasars, prefers a dark energy component whose equation of state is dynamical. In the simplest two-parameter description of this behavior, the Chevallier–Polarski–Linder (CPL) parametrization, the data favor the quadrant $w_0 > -1$, $w_a < 0$, which implies that the equation of state crosses the phantom divide $w = -1$ at a redshift $z \simeq 0.4$. We show that the local limit of nonlocal gravity, a teleparallel extension of general relativity governed by a scalar function called gravitational susceptibility $S(x)$, offers a natural explanation. In this model a cosmological constant is not allowed; dark energy is necessarily dynamical. The phantom crossing seen by an observer is an effective phenomenon generated by the time evolution of $S(z)$. We establish a correspondence: to every general-relativistic dark energy model with equation of state $w(z)$, there corresponds a modified TEGR cosmology with $w = -1$ and a suitable susceptibility $S(z)$ that are indistinguishable at the level of the background expansion. We construct the susceptibility functions that reproduce the CPL behavior favored by DESI data, and we confront the model with the combined CMB,BAO, and SNe data using Boltzmann solver codes. The recovered susceptibility displays a characteristic low-redshift dip with depth $\beta = -0.025 \pm 0.007$ at $z \simeq 0.4$, and its derivative changes sign at a redshift equal to the phantom-crossing epoch.

## I. INTRODUCTION

The standard cosmological model, spatially flat ΛCDM, which combines a cosmological constant Λ with cold dark matter (CDM), successfully explains a wide range of observations, most notably the temperature and polarization anisotropies of the cosmic microwave background (CMB) [1]. Yet the physical nature of dark energy (DE) and dark matter (DM) remains unknown, and the cosmological constant itself suffers from severe theoretical problems [2, 3]. It is therefore not surprising that the search for deviations from cosmological constant solution for an accelerating universe has become one of the main endeavors of observational cosmology today [4, 5].

The Dark Energy Spectroscopic Instrument (DESI) was designed precisely for this purpose [6]. It measures the baryonic acoustic oscillation (BAO) through the large scale structure survey. The BAO data alone are described well by a flat ΛCDM model, but the parameters preferred by BAO are in mild tension with those determined from the CMB: the ΛCDM fit to DESI+CMB shows a $2.3\sigma$ tension. This tension disappears when the dark energy equation of state is allowed to vary with time [7–9].

The statistical preference for dynamical dark energy, and in particular the exact significance, depends on the data combination and on the assumed parametrization [10–15]. What matters for the present paper is that the data consistently point toward a dark energy equation of state that is dynamical and crosses $w = -1$ at low redshift, known as phantom crossing, and that this pattern is difficult to realize in general relativity with a single field [16–18]. A phantom component is problematic in general relativity (GR) because it violates the null energy condition [19, 20]. Modified gravity offers a way out: the crossing may be an effective description of a modified expansion history rather than a property of the dark energy fluid itself [21–27]. Various proposals have been put forward to accommodate the crossing, ranging from multi-field models [16, 28] to modified-gravity constructions in which the crossing is only an effective feature [29].

In this paper we show that the local limit of nonlocal gravity (NLG) provides a well-motivated gravitational framework in which the phantom crossing appears naturally as an effective phenomenon. NLG is a classical, nonlocal generalization of GR inspired by the nonlocal electrodynamics of media [30–32]. This theory is formulated as a modification in the teleparallel gravity framework, and its local limit, obtained when the nonlocal kernel collapses to a delta function, resulting in the modified teleparallel equivalent of general relativity (modified TEGR, or MTEGR) model that differs from the base TEGR by a dimensionless scalar function $S(x)$, known as the gravitational susceptibility [33].

The cosmological consequences of MTEGR were studied extensively in [33]. One of the important results of this study is that out of different standard FLRW models, only the modified Cartesian flat model is consistent with a time-dependent susceptibility function, and in this model de Sitter spacetime is not a valid solution. A cosmological constant is therefore excluded; instead, dark energy must be dynamical. The modified continuity equation of this model implies that the energy density of each component scales with the susceptibility, enforcing dynamics on every component, including dark energy (with, for example, $w = -1$) [33, 34]. The model was shown to be capable of improving the $H_0$ tension [33, 35, 36], and its linear perturbations were studied and tested with the CMB

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data in [37] for tighter constraints.

An observer living in the MTEGR universe with a constant DE pressure to density ratio (e.g. $w = -1$), who interprets the data with the standard Friedmann equations, may infer the effects of time evolution of $S(z)$ on all densities as a dynamical equation of state for DE. This is the key observation that motivates the present work. We show that the resulting effective equation of state $\bar{w}_{\rm de}(z)$ can cross $-1$ even though the true dark energy component has fixed equation of state ($w = -1$) at all times. The crossing is therefore not a property of the dark energy fluid itself, but a consequence of the modified gravitational framework. Given any DE model with known equation of state $w(z)$ within general relativity, we find the susceptibility $S(z)$ that reproduces it at the background cosmology level. Applying this procedure to the CPL model preferred by DESI DR2, we obtain the required $S(z)$. The background analysis gives us a baseline ansatz for $S(z)$, that proves to be helpful in a full (including perturbation) cosmological analysis. Using Monte Carlo Markov Chain (MCMC) analysis, with the CMB, the DESI DR2 BAO data and different supernovae catalogs, we constrain the ansatz parameters and approve that the model is preferred according to the reported Akaike information criterion (AIC), and deviance information criterion (DIC) [38, 39] by the data, over the standard cosmology model.

The structure of this paper is as follows. In Section II, we review the results of DESI survey. In Section III, we summarize the local limit of nonlocal gravity. In Section IV we derive the effective dark energy density, pressure and equation of state inferred by an observer in this model, and establish the correspondence between $w(z)$ models and $S(z)$ models. In Section V we set up the phenomenological ansatz for $S(z)$ guided by the CPL target. Section VI describes the data sets and the numerical tools. Section VII contains the results of the analysis, and Section VIII discusses the physical implications of the phantom crossing within this framework and we present our conclusion.

## II. DESI DR2 AND PHANTOM CROSSING

The Dark Energy Spectroscopic Instrument (DESI) was designed for the purpose of detecting the deviations of cosmological expansion from the standard ΛCDM model [6]. Its second data release (DR2) [7] contains spectra of more than fourteen million galaxies and quasars. From this sample, the collaboration extracted baryon acoustic oscillation measurements in a number of redshift bins spanning the range $0.1 \lesssim z \lesssim 4.2$, with the highest-redshift bins coming from the Lyman-$\alpha$ forest [40, 41]. The DR2 BAO data are consistent with the DR1 measurements and with the earlier SDSS results, and the distance-redshift relation they trace agrees with that of the recent supernova compilations over the same redshift range [7].

The combined DESI+CMB data exclude ΛCDM at $3.1\sigma$, and adding supernovae raises the significance to $2.8$–$4.2\sigma$ depending on the sample used.

The most striking feature of the DESI DR2 results is that the preferred dynamical dark energy crosses the phantom divide. The simplest and most widely used description of a time-varying dark energy equation of state is the phenomenological Chevallier–Polarski–Linder (CPL) parametrization [42, 43],

$$w(a) = w_0 + w_a(1-a)\,, \tag{1}$$

or, equivalently, $w(z) = w_0 + w_a z/(1+z)$. For the combination of DESI BAO, CMB and the DES Year-5 supernovae (DESY5) [44], the DESI collaboration reports [7]

$$w_0 = -0.752 \pm 0.057\,, \qquad w_a = -0.86^{+0.23}_{-0.20}\,, \tag{2}$$

which deviates up to $4.2\sigma$ from base ΛCDM. These numbers are consistent with, and sharper than, the DR1 findings, for which the corresponding preference was $2.6\sigma$ with CMB alone and up to $3.9\sigma$ with supernovae [45].

The crucial feature of the result in equation (2) is the sign pattern $w_0 > -1$, $w_a < 0$. Using equations (1) and (2), one finds that the equation of state at the present epoch is quintessence-like, $w(z=0) = w_0 > -1$, while $w(z) = w_0 + w_a z/(1+z)$ decreases with redshift and crosses the phantom divide, $w = -1$, at

$$z_\times = \frac{-(1+w_0)}{1+w_0+w_a} \simeq 0.405\,. \tag{3}$$

For $z > z_\times$ the preferred dark energy is phantom, $w < -1$. This crossing is one of the important features of the DESI DR2 data [7].

Alcock-Paczyński Measurements from the Lyman-$\alpha$ Forest in a recent study put tighter constraints on the dynamics of DE in the matter-dominated era as well [41].

## III. LOCAL LIMIT OF NONLOCAL GRAVITY

In this section we summarize the fundamentals of the local limit of nonlocal gravity, which is the theoretical framework underlying our analysis. We begin by reviewing how the nonlocal constitutive relation of NLG reduces to a local one governed by a single scalar susceptibility $S(x)$. We then introduce the modified Cartesian flat cosmology as a solution to the field equations of this model in the case of a time-dependent $S$.

### A. From nonlocal gravity to modified TEGR

Nonlocal gravity is a classical extension of GR constructed in analogy with the nonlocal electrodynamics of media [30–32]. The theory is formulated in the teleparallel gravity framework, in which spacetime is parallelized by a preferred tetrad frame field $e^{\mu}{}_{\hat{\alpha}}(x)$, and the gravitational field strength is

the torsion tensor $C_{\mu\nu\rho}$ and contorsion tensor $K_{\mu\nu\rho}$ of the Weitzenböck connection, defined as

$$C_{\mu\nu}{}^{\alpha} = e^{\alpha}{}_{\hat{\beta}} \left( \partial_{\mu} e_{\nu}{}^{\hat{\beta}} - \partial_{\nu} e_{\mu}{}^{\hat{\beta}} \right) , \tag{4}$$

$$K_{\mu\nu\rho} = \frac{1}{2} \left( C_{\mu\rho\nu} + C_{\nu\rho\mu} - C_{\mu\nu\rho} \right) . \tag{5}$$

GR, when written in this language, becomes the teleparallel equivalent of general relativity (TEGR), and the Einstein equation is rewritten in a new form,

$$\frac{\partial}{\partial x^{\nu}} \mathfrak{H}^{\mu\nu}{}_{\hat{\alpha}} + \frac{\sqrt{-g}}{8\pi G} \Lambda \, e^{\mu}{}_{\hat{\alpha}} = \sqrt{-g} \left( T_{\hat{\alpha}}{}^{\mu} + \mathbb{T}_{\hat{\alpha}}{}^{\mu} \right) , \tag{6}$$

in which the gravitational field tensor $\mathfrak{H}_{\mu\nu\rho}$ is related to the auxiliary torsion tensor, $\mathfrak{C}_{\alpha\beta\gamma} := C_{\alpha}\, g_{\beta\gamma} - C_{\beta}\, g_{\alpha\gamma} + K_{\gamma\alpha\beta}$, by a simple, local constitutive relation

$$\mathfrak{H}_{\mu\nu\rho} := \frac{\sqrt{-g}}{8\pi G} \, \mathfrak{C}_{\mu\nu\rho} . \tag{7}$$

This is in close analogy with $(\mathbf{D}, \mathbf{H})$ versus $(\mathbf{E}, \mathbf{B})$ in electrodynamics. The term $\mathbb{T}_{\hat{\alpha}}{}^{\mu}$ in equation (6) is the energy-momentum corresponding to the gravitational field, defined as

$$\mathbb{T}_{\mu\nu} = (\sqrt{-g})^{-1} \left( C_{\mu\rho\sigma} \mathfrak{H}_{\nu}{}^{\rho\sigma} - \frac{1}{4} g_{\mu\nu} C_{\rho\sigma\delta} \mathfrak{H}^{\rho\sigma\delta} \right) . \tag{8}$$

In NLG, the constitutive relation (equation (7)) is modified to become nonlocal: the excitation at an event receives contributions from the past history of the field, weighted by a causal kernel $\mathcal{K}(x, x')$ [32, 46],

$$\mathfrak{H}_{\mu\nu\rho} = \frac{\sqrt{-g}}{8\pi G} (\mathfrak{C}_{\mu\nu\rho} + N_{\mu\nu\rho}) , \tag{9}$$

$$N_{\hat{\mu}\hat{\nu}\hat{\rho}}(x) := \int \mathcal{K}(x, x') \, \mathfrak{C}_{\hat{\mu}\hat{\nu}\hat{\rho}}(x') \sqrt{-g(x')} \, d^4x' . \tag{10}$$

Using this ansatz and substituting $\mathcal{H}_{\mu\nu\rho}$ back into the TEGR field equations (equation (6)), we obtain a set of integro-differential equations that can be solved in various simplifying regimes to extract physical features and connect to phenomenology.

The local limit of this nonlocal relation is obtained by taking the kernel to be proportional to a four-dimensional Dirac delta function,

$$\mathcal{K}(x, x') = \frac{S(x)}{\sqrt{-g(x)}} \, \delta(x - x') , \tag{11}$$

where $S(x)$ is a dimensionless scalar function [33]. In this limit the constitutive relation becomes

$$N_{\mu\nu\rho} = S(x) \, \mathfrak{C}_{\mu\nu\rho}(x) . \tag{12}$$

The resulting theory, modified TEGR (MTEGR), is TEGR extended by a single scalar function $S(x)$, the gravitational susceptibility, which is characteristic of the background spacetime in the same sense that the electric permittivity and magnetic permeability are characteristic of a medium in electrodynamics. For $S = 0$ one recovers TEGR and hence general relativity. The attractive nature of gravitational force implies $1 + S > 0$. There is no field equation for $S(x)$; like the constitutive functions of a medium, it must be determined from observation, see [33, 47].

### B. The modified Cartesian flat model

A spatially flat FLRW universe, whose tetrad frame is Cartesian ($e_{\mu}{}^{\hat{\alpha}} = a(\eta)\delta^{\alpha}_{\mu}$ , $e^{\mu}{}_{\hat{\alpha}} = a^{-1}(\eta)\delta^{\mu}_{\alpha}$), is a solution of the MTEGR field equations that we will use as a basis for our cosmological analysis. For any other FLRW representation, the field equations force $S$ to be constant. The Cartesian flat case allows for a time-dependent susceptibility, which is the natural companion of a dynamical background [33, 48].

The background dynamics of this model is governed by the modified Friedmann equations, derived from the modified field equations of MTEGR. With cosmic time $t$, scale factor $a(t)$, Hubble parameter $H = \dot{a}/a$, and the total energy density and pressure of matter, radiation and dark energy, $\rho = \sum_i \rho_i$ and $P = \sum_i P_i$, one finds [33]

$$3(1 + S)H^2 = \Lambda + 8\pi G \, \rho , \tag{13}$$

$$2(1 + S)\frac{\ddot{a}}{a} + (1 + S)H^2 = \Lambda - 8\pi G \, P - 2\frac{dS}{dt} H . \tag{14}$$

For $S = 0$, these reduce to the standard Friedmann equations. Two properties of equations (13) and (14) are crucial for what follows. First, de Sitter spacetime is not a solution of the modified Cartesian flat model: setting $\rho = P = 0$ in equations (13) and (14) leads to a contradiction unless $dS/dt = 0$ [33, 46]. A cosmological constant is therefore excluded, and dark energy must be a dynamical component. There is more detail on this subject in Appendix A. Second, the susceptibility enters the dynamics of every component: differentiating equation (13) and using equation (14), one obtains the modified continuity equation ($\Lambda = 0$) [33]

$$\frac{d\rho_i}{dt} + 3H(\rho_i + P_i) + \frac{\dot{S}}{1 + S} \rho_i = 0 , \tag{15}$$

which is assumed to hold separately for each component $i$ with $P_i = w_i \rho_i$. Its solution is

$$\rho_i(t) = \rho_i(t_0) \, \frac{1 + S(t_0)}{1 + S(t)} \, a(t)^{-3(1+w_i)} , \tag{16}$$

where $t_0$ denotes the present epoch ($a(t_0) = 1$). The factor $[1 + S(t_0)]/[1 + S(t)]$ is the imprint of this modified gravity model on every density. In particular, for a dark energy component with $w = -1$,

$$\rho_{\rm de}(t) = \rho_{\rm de}(t_0) \, \frac{1 + S(t_0)}{1 + S(t)} , \tag{17}$$

so that even the "cosmological-constant-like" component is dynamical: its density follows the inverse of $1 + S(t)$ and is constant only if $S$ is constant [33, 34]. This is the origin of

dynamical dark energy in MTEGR, but it does not yet imply a dynamical equation of state. To achieve that and allow for phantom crossing, we will adopt a different approach in subsequent chapters.

### C. Cosmological perturbations

A full confrontation of the model with the CMB data requires the linear perturbation equations about the modified Cartesian flat background. Here we collect the essential ingredients, and refer the reader to Reference [37] for the complete derivation and for the synchronous-gauge equations that are actually implemented in the Boltzmann code. Because MTEGR is a tetrad theory, one perturbs the frame field rather than the metric,

$$e_\mu{}^{\hat{\alpha}}(x) = a(\eta)\big[\delta_\mu^\alpha + \psi_\mu{}^{\hat{\alpha}}(x)\big]\,,$$
$$e^\mu{}_{\hat{\alpha}}(x) = a^{-1}(\eta)\big[\delta_\alpha^\mu - \psi^\mu{}_{\hat{\alpha}}(x)\big]\,, \tag{18}$$

where $\eta$ is conformal time and $\psi_\mu{}^{\hat{\alpha}}$ is treated to linear order away from the Minkowski background. The susceptibility is likewise split into a time-dependant background and an inhomogeneous perturbation,

$$S(\eta,\mathbf{x}) = \bar{S}(\eta) + \delta S(\eta,\mathbf{x})\,. \tag{19}$$

Adopting the conformal Newtonian gauge and the standard scalar–vector–tensor decomposition of the tetrad perturbation [49, 50], the scalar sector is described by the three gravitational potentials $\phi$, $\zeta$, and $\psi$ together with $\delta S$. These scalar fields are related to the tetrad perturbation via the relation

$$\psi_{00} = \psi\,, \qquad \psi_{0i} = -\psi_{i0} = \partial_i\zeta\,, \qquad \psi_{ij} = \delta_{ij}\,\phi\,. \tag{20}$$

Using the orthonormality condition of the tetrads, $e_\mu{}^{\hat{\alpha}} e_\nu{}^{\hat{\beta}} \eta_{\hat{\alpha}\hat{\beta}} = g_{\mu\nu}$, we can connect the metric and tetrad perturbations together. Here the nonzero components of metric perturbation are

$$g_{\mu\nu} = -a^2(\eta)(1-2\psi)\,, \quad g_{ij} = a^2(\eta)[(1+2\phi)\delta_{ij}]\,. \tag{21}$$

The field equations of MTEGR tie these quantities together algebraically,

$$\phi - \psi + ß\,\zeta = 0\,, \qquad \mathcal{H}\,\delta S = \frac{d\bar{S}}{d\eta}\,\phi\,, \tag{22}$$

where $\mathcal{H} = \frac{da}{d\eta}/a$ and we have introduced the shorthand $ß \equiv \frac{d\bar{S}}{d\eta}/(1+\bar{S})$. The first relation shows that the gravitational slip is sourced by the susceptibility through the nondynamical field $\zeta$, while the second links the perturbation of the medium to the Newtonian potential. The scalar potentials obey the first-order evolution equation

$$\frac{d\phi}{d\eta} + \Big(-\frac{1}{3\mathcal{H}}\Delta + \mathcal{H} + \frac{1}{2}ß\Big)\phi = \frac{\kappa\,a^2}{6(1+\bar{S})\mathcal{H}}\,\delta\rho\,. \tag{23}$$

The scalar sector is completed by the modified Poisson equation, which sources the Newtonian potential with the matter inhomogeneity,

$$-\Delta\phi + 3\mathcal{H}\big(\frac{d\phi}{d\eta} + \mathcal{H}\psi\big) = \frac{\kappa\,a^2\,\delta\rho - 3\mathcal{H}^2\,\delta S}{2(1+\bar{S})}\,, \tag{24}$$

which reduces to the familiar $\Delta\phi = 4\pi G\,a^2\,\delta\rho$ in the general-relativistic limit $ß = 0$ [51].

Every modification of the perturbation equations is controlled by the single background function $ß(\eta)$, so that once the ansatz for $S(z)$ is fixed, the dynamics of the perturbations is fully specified and the model contains no new free functions or parameters beyond those already present at the background level. We solve the coupled system above in the synchronous gauge [52, 53] inside a modified version of the Boltzmann code `CLASS` (Section VI), which supplies the CMB angular power spectra used in our MCMC analysis.

### D. Newtonian limit

In the Newtonian regime, where the susceptibility is constant over the local environment ($S(x) = \mathbb{S}$), the modified field equations yield the equation

$$\nabla^2\Phi = \frac{4\pi G\,\rho}{1+\mathbb{S}}\,, \tag{25}$$

which is equivalent to Newtonian gravity with $G \to G/(1+\mathbb{S})$.

Local measurements of the gravitational constant do not constrain $\mathbb{S}$ directly [33]. Because the bare gravitational constant $G$ is a free parameter, experiments measure the degenerate combination

$$G_{\rm obs} \equiv \frac{G}{1+\mathbb{S}}\,, \tag{26}$$

and any constant value of $\mathbb{S}$ can be absorbed into a redefinition of $G$. Thus, as long as the susceptibility is constant locally, solar system physics places no bound on its magnitude.

The situation changes when $S(z)$ evolves cosmologically. A time-dependent susceptibility introduces $\dot{S}$ terms into the cosmological equations, breaking the static degeneracy. The condition $S(0) \simeq 0$ adopted in Section V is therefore a normalization choice: it fixes the bare $G$ such that the present-day cosmological gravitational strength matches the locally measured $G_{\rm obs}$. With this normalization, the low-redshift variation of $S(z)$ that drives phantom crossing is confined to a genuine large-scale cosmological effect, safely decoupled from local physics.

We note that in the Newtonian regime of the parent nonlocal gravity theory, the nonlocal kernel generates an effective dark matter component that explains galaxy rotation curves [54]; in the local limit, this halo collapses onto the source (due to local kernel $K(x,x') \sim \delta(x-x')$), so MTEGR itself is not a dark matter substitute at the galactic scale [33].

## IV. EFFECTIVE DARK ENERGY

In this section, we use a background-level analysis to establish an effective connection between dynamical dark energy within the standard gravitational framework of cosmology and the cosmology of the MTEGR model, with a specific time evolution of the susceptibility function.

### A. The effective dark energy density

An observer in the MTEGR universe measures distances, redshifts, and the expansion rate $H(z)$. All cosmological observables are ultimately functions of $H(z)$ through the comoving distance

$$r(z) = \int_0^z \frac{dz'}{H(z')}\,, \tag{27}$$

from which angular diameter distances $D_A(z) = r(z)/(1+z)$ and luminosity distances $D_L(z) = (1+z)r(z)$ are derived for SNe Ia and BAO analyses, respectively. The comparison between the model and data is therefore made through the modified Friedmann equation (equation (13)). With $1+z = 1/a$ and $S(t) = S(z)$, equation (16) gives

$$\rho_i(z) = \rho_{i0}\,\frac{1+S(0)}{1+S(z)}\,(1+z)^{3(1+w_i)}\,, \tag{28}$$

and equation (13) becomes

$$H^2(z) = \frac{8\pi G}{3}\,\frac{1+S(0)}{\big[1+S(z)\big]^2}\,\big[(1+z)^3\rho_{m0} + \rho_{\rm de0}\big]\,, \tag{29}$$

where we have neglected relativistic components, which are irrelevant at the redshifts of interest here. Defining the critical density by $\rho_c = \rho_{m0} + \rho_{\rm de0}$, and $\Omega_i = \rho_{i0}/\rho_c$, we obtain

$$H^2(z) = H_0^2\left[\frac{1+S(0)}{1+S(z)}\right]^2\big[(1+z)^3\Omega_{m0} + \Omega_{\rm de0}\big]\,, \tag{30}$$

where we have used $H_0^2 = 8\pi G\rho_c/[3(1+S(0))]$.

Equation (30) is the expansion history of the modified model. Let us now ask how this history would be *interpreted* by a standard observer who assumes general relativity. Such an observer would parameterize the same $H(z)$ as

$$H^2(z) = H_0^2\left[(1+z)^3\Omega_m^{\rm eff} + \Omega_{\rm de}^{\rm eff}(z)\right], \tag{31}$$

where $\Omega_m^{\rm eff}$ is the effective matter density parameter inferred from the data under the GR assumption, and $\Omega_{\rm de}^{\rm eff}(z)$ is the effective dark energy density that would be required to match the observed $H(z)$.

Crucially, for an observer analyzing actual data, $\Omega_m^{\rm eff}$ is not a free convention; it is inferred from observables that are sensitive to the combination $\Omega_m H_0^2$. The BAO scale provides the strongest such constraint. In the GR framework, the sound horizon at the drag epoch,

$$r_d = \int_{z_d}^{\infty} \frac{c_s(z)}{H(z)}dz, \tag{32}$$

is fixed by the CMB and provides a standard ruler. BAO measurements of the transverse and radial scales,

$$D_A(z)/r_d \quad \text{and} \quad H(z)r_d, \tag{33}$$

directly constrain the combination $\Omega_m^{\rm eff}h^2$ (where $h = H_0/(100\,{\rm km\,s^{-1}Mpc^{-1}})$) through the shape of the matter power spectrum. Similarly, SNe Ia provide a model-independent measurement of $D_L(z)$ and therefore of the integrated expansion history, which together with BAO breaks degeneracies and yields a best-fit value for $\Omega_m^{\rm eff}$.

Equating the physical $H(z)$ from the modified theory (equation (30)) with the GR-interpreted form (equation (31)), we obtain the effective dark energy density that the observer would infer

$$\Omega_{\rm de}^{\rm eff}(z) = -\Omega_m^{\rm eff}(1+z)^3 + \left[\frac{1+S(0)}{1+S(z)}\right]^2\big[(1+z)^3\Omega_{m0} + \Omega_{\rm de0}\big]\,. \tag{34}$$

At this point, the physical interpretation of $\Omega_m^{\rm eff}$ must be fixed. For a standard observer, $\Omega_m^{\rm eff}$ is not arbitrary: it is the value that simultaneously fits the BAO power spectrum shape (which constrains $\Omega_m^{\rm eff}h^2$) and the CMB acoustic scale. In the modified theory, the true matter density today is $\Omega_{m0}$, and the susceptibility modifies the effective gravitational coupling, thereby altering the matter power spectrum and the inferred value of $\Omega_m$. A natural and physically motivated choice is

$$\Omega_m^{\rm eff} = \frac{\Omega_{m0}}{1+S(0)}, \tag{35}$$

which ensures that the effective matter density today, as inferred from the $z = 0$ limit of the Friedmann equation, matches the true density rescaled by the present-day susceptibility. This choice is equivalent to identifying the combination $8\pi G\rho_{m0}/[1+S(0)]$ as the observed gravitational source density, consistent with the Poisson equation in the Newtonian limit. With this identification, equation (34) becomes

$$\Omega_{\rm de}^{\rm eff}(z) = \frac{1+S(0)}{\big[1+S(z)\big]^2}\,\Omega_{\rm de0} + \left[\left(\frac{1+S(0)}{1+S(z)}\right)^2 - 1\right](1+z)^3\,\Omega_m^{\rm eff}\,. \tag{36}$$

At $z = 0$, this reduces to

$$\Omega_{\rm de}^{\rm eff}(0) = \frac{\Omega_{\rm de0}}{1+S(0)}, \tag{37}$$

which is the natural counterpart of equation (35): the effective present dark energy density is the true one rescaled by $1 + S(0)$.

Equation (36) is the central object of this section. It shows that the dark energy density inferred by an observer, assuming GR, from distance measurements (SNe Ia, BAO) differs

from the true dark energy density of the modified model. The difference is controlled entirely by the susceptibility $S(z)$ and by the effective matter density $\Omega_m^{\rm eff}$ that the observer would infer from BAO and CMB data. In practice, this means that a time-varying $S(z)$ can produce an apparent evolution of the dark energy equation of state, potentially mimicking phantom behavior, while the underlying model remains consistent with the measured distance-redshift relation and the BAO-standard-ruler constraints.

Importantly, the success of this effective description hinges on whether the modified model can simultaneously fit the primary cosmological probes. Any deviation from a cosmological constant, including phantom crossing, is then an observational signature of the underlying modified gravity theory, encoded in $S(z)$ and in the mismatch between the true and inferred matter densities.

### B. The effective pressure and the effective equation of state

The second Friedmann equation (equation (14)) determines the acceleration of the universe, and with it the effective pressure of dark energy. Setting $P_{\rm de} = -\rho_{\rm de}$ ($w = -1$) in equation (14), and expressing the result in terms of redshift through

$$\frac{d\ln(1+S)}{dt} = -(1+z)H\,\frac{d\ln(1+S)}{dz}\,, \tag{38}$$

we find, after using equation (13) to eliminate $H^2$,

$$\bar{P}_{\rm de}(z) = -\frac{\rho_{\rm de0}}{1+S(z)} - \tag{39}$$
$$\frac{2}{3}\,\frac{d\ln(1+S)}{dz}\,\frac{1+S(0)}{\left[1+S(z)\right]^2}\,(1+z)\left[(1+z)^3\rho_{m0}+\rho_{\rm de0}\right].$$

Here $\bar{P}_{\rm de}$ is the pressure that an observer using the standard equations would assign to the dark energy in order to reproduce the measured acceleration.

The effective equation of state is the ratio of the two,

$$\bar{w}_{\rm de}(z) = \frac{\bar{P}_{\rm de}(z)}{\bar{\rho}_{\rm de}(z)}\,. \tag{40}$$

Before evaluating it, it is convenient to introduce a normalized version of the susceptibility. Define

$$\mathfrak{S}(z) := \frac{1+S(z)}{1+S(0)}\,, \tag{41}$$

so that $\mathfrak{S}(0) = 1$. In terms of $\mathfrak{S}$ and of the ratio

$$\chi := \frac{\rho_{m0}}{\rho_{\rm de0}}\,, \tag{42}$$

the effective equation of state (equation (40)) takes the compact form

$$\bar{w}_{\rm de}(z) = \frac{-\mathfrak{S}(z) - \dfrac{2}{3}\,\dfrac{d\ln\mathfrak{S}}{dz}\,(1+z)\left[(1+z)^3\chi+1\right]}{1+\left[1-\mathfrak{S}^2(z)\right](1+z)^3\,\chi}\,. \tag{43}$$

This is the main result of this section. Its structure is simple and revealing: If $S$ is constant, then $\mathfrak{S} = 1$ and equation (43) reduces to $\bar{w}_{\rm de} = -1$: the effective dark energy is a cosmological constant, as expected. The deviation from $-1$ is controlled by two terms: the algebraic term $1-\mathfrak{S}^2$ in the denominator, and the derivative term $d\ln\mathfrak{S}/dz$ in the numerator, which involves the rate of change of the susceptibility. The derivative term in the numerator carries a minus sign with respect to $d\ln\mathfrak{S}/dz$: a decreasing susceptibility ($d\ln\mathfrak{S}/dz < 0$) makes the numerator less negative and pushes $\bar{w}_{\rm de}$ above $-1$ (the quintessence side), while an increasing susceptibility ($d\ln\mathfrak{S}/dz > 0$) makes it more negative and pushes $\bar{w}_{\rm de}$ *below* $-1$, toward the phantom region.

The phantom crossing therefore occurs near the susceptibility's minimum, meaning when $d\ln\mathfrak{S}/dz$ changes sign from negative to positive, provided the terms are large enough. In the dip picture of Section V, the crossing sits in the vicinity of the bottom of the dip, on its rising side: as $z$ increases, $\mathfrak{S}$ first decreases (driving $\bar{w}_{\rm de}$ above $-1$), reaches its minimum, and then increases again (driving $\bar{w}_{\rm de}$ below $-1$). This is the mechanism we will exploit.

### C. Correspondence between $w(z)$ models and $S(z)$ models

Equation (43) establishes a correspondence at the level of the background, therefore we claim that for every cosmological model based on general relativity in which dark energy has an equation of state $w(z)$, there exists a cosmology based on MTEGR, with a suitable susceptibility $S(z)$ and a dark energy component with $w = -1$, such that the two models are indistinguishable at the level of the background expansion.

The claim follows directly from equation (43): demanding $\bar{w}_{\rm de}(z) = w(z)$ turns this equation into a first-order ordinary differential equation for $\mathfrak{S}(z)$,

$$w(z) = \frac{-\mathfrak{S} - \dfrac{2}{3}\,\dfrac{d\ln\mathfrak{S}}{dz}\,(1+z)\left[(1+z)^3\chi+1\right]}{1+\left[1-\mathfrak{S}^2\right](1+z)^3\,\chi}\,, \tag{44}$$

which can be solved numerically from $z = 0$ with the initial condition $\mathfrak{S}(0) = 1$, for any prescribed $w(z)$ and any fixed $\chi > 0$. The resulting $\mathfrak{S}(z)$ reproduces the given $w(z)$ by construction. For a fixed $\chi$ value (fixed, for example, by the local matter abundance) the mapping is deterministic. Equation (44) has no closed-form solution in general, but it is trivially integrated numerically. We emphasize that the dark energy component of the modified model never leaves $w = -1$; the dynamics of the effective equation of state is entirely a gravitational effect, encoded in $\mathfrak{S}(z)$.

This correspondence is the theoretical backbone of the present paper. It tells us that the DESI DR2 preference for a crossing CPL equation of state does not, by itself, require phantom matter or any exotic fluid: it can be interpreted as a measurement of the susceptibility function $S(z)$ of the local limit of nonlocal gravity.

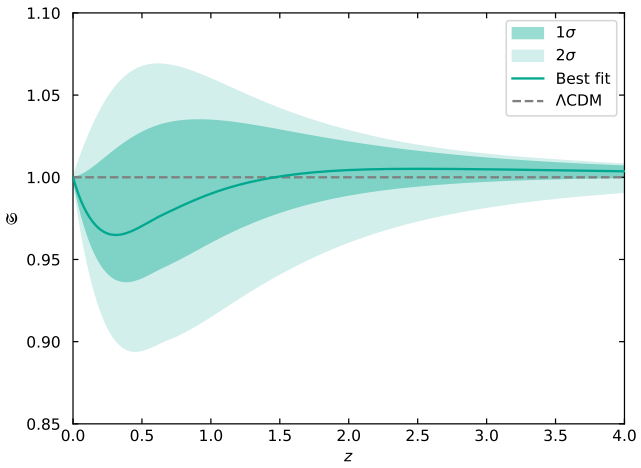


FIG. 1: The susceptibility $\mathfrak{S}(z)$ from equation (44) for CPL parameters (equation (2)) with $\chi = 0.45$. The solid line is the CMB+BAO+DESY5 best fit; the colored regions are the $1\sigma$ and $2\sigma$ confidence levels of $(w_0, w_a)$. The DESI DR2 sample was used for the plot.

## V. DYNAMIC DARK ENERGY ANSATZ

In this section, we take a first step toward the numerical analysis by choosing an appropriate ansatz for $S(z)$ based on the evidence presented thus far.

### A. The CPL target

Guided by the DESI DR2 analysis, which focuses on the CPL parametrization, we now use equation (44) with

$$w(z) = w_0 + w_a \frac{z}{1+z}\,, \tag{45}$$

and with the best-fit values of equation (2), namely $w_0 = -0.752$, $w_a = -0.86$. We fix $\chi = 0.45$, consistent with the standard cosmological determination of the matter-to-dark-energy ratio, and integrate equation (44) from $z = 0$ with $\mathfrak{S}(0) = 1$ toward large redshifts. The result is shown in Fig. 1. The figure displays the distribution of $\mathfrak{S}(z)$ curves obtained by sampling the CPL parameters $(w_0, w_a)$ within their $1\sigma$ and $2\sigma$ confidence regions, while the curve computed with the exact best-fit values in equation (2) is highlighted as a solid line. The spread of this family shows how the observational uncertainty on the equation of state translates into an uncertainty in the parameters controlling the shape of $\mathfrak{S}$.

Two lessons follow from this exercise. First, the sign change of $\mathfrak{S}'(z)$ and hence of $S'(z)$ occurs at a redshift near the phantom-crossing redshift of the CPL model. Second, the dip in $\mathfrak{S}(z)$ is what turns, through the derivative term in equation (43), into the phantom crossing of the effective equation of state. This is consistent with the analytic structure of equation (43) discussed above. Although this background-level reconstruction is not by itself a full cosmological analysis, it tells us what the susceptibility must look like in order to explain the DESI DR2 data: it must have a well-defined dip in the low-redshift region $z \lesssim 1$, and it must return to a constant (in practice, to $S(0)$) at higher redshifts, so that the early universe is unaffected.

### B. Choosing the functional form of $S(z)$

The reconstruction above suggests a phenomenological ansatz for $S(z)$ with the following properties: (i) a dip at $z \lesssim 1$, (ii) recovery to a constant at large $z$, and (iii) as few parameters as possible. A further simplification is available: the analysis of the modified model reveals a degeneracy between $H_0$ and $1 + S(0)$ (or as previously discussed between $G$ and $1 + S(0)$), the two quantities always appear in combination, so that we may fix $S(0) = 0$ without loss of generality. With this choice, $\mathfrak{S}(z) = 1 + S(z)$.

Our ansatz is a single, fully analytic function, that begins with an additional simplifying assumption: $\mathfrak{S}(\infty) = \mathfrak{S}(0) = 1$, which reduces the generality here, but many unreleased numerical experiments strongly support it. We therefore propose

$$\mathfrak{S}(z) = 1 + \left(\alpha z + \beta z^2\right) e^{-z/\delta}\,. \tag{46}$$

This form is continuous and differentiable to all orders, contains the same number of additional parameters as the CPL model, and automatically satisfies $\mathfrak{S}(0) = \mathfrak{S}(\infty) = 1$. In practice, the parameter $\alpha$ turns out to be strongly constrained to values very close to zero by the data, so the effective form used in the final analysis, after a reparametrization, and by setting $\alpha = 0$, is the two-parameter function

$$\mathfrak{S}(z) = 1 + \frac{\beta}{\delta^2} z^2 e^{2(1-\frac{z}{\delta})}\,. \tag{47}$$

This form ensures that the parameter $\delta$ independently determines the dip's position, and $\beta$ determines the dip's height. Note that with $S(0) = 0$ and $\mathfrak{S} = 1 + S$, the condition $1+S > 0$ requires $\mathfrak{S}(z) > 0$ for all $z$, which is easily satisfied by this ansatz if $\beta > -1$. It is also easy to see that $\delta > 0$ is a reasonable prior choice, and that a dip would only occur if $\beta < 0$. Although we do not enforce this negativity as a prior on $\beta$ in our numerical analysis, the resulting posterior for $\beta$ nevertheless satisfies $\beta < 0$. It is also evident that, for small $|\beta|$, the parameter $\delta$ becomes ineffective and has negligible impact on the physics. Consequently, we expect the posterior distributions of $\delta$ to broaden near the $\beta = 0$ line.

## VI. DATA AND METHODS

This section introduces and reviews the tools, numerical methods, and data sets used in the analysis.

### A. Data sets

The analysis is based on different data-set permutations that, according to the DESI DR2 collaboration, most strongly support dynamical dark energy [7]. Each of these sets plays a special role in constraining the cosmological parameters, and we explore multiple supernova catalogs to assess the robustness of our results.

- **CMB.** We use the final Planck data release (PR4), based on the NPIPE processing pipeline [1]. Specifically, we use the full temperature and polarization likelihood, including the $TT$, $TE$, and $EE$ power spectra and their covariance matrices. We refer to this combination collectively as $TTTEEE$.
- **BAO.** We use the thirteen BAO measurements from the second DESI data release (DR2) [7], covering galaxy, quasar, and Lyman-$\alpha$ tracers. These constitute the most precise BAO measurements available to date.
- **Supernovae.** To test the sensitivity of our results to the choice of supernova catalogue, we perform the analysis using three distinct samples: the Dark Energy Survey Year-5 (DESY5) sample [44], the Pantheon+ compilation [55], and the Union3 sample [56]. As reported by the DESI collaboration, the combination involving the DESY5 sample yields the strongest preference for a dynamical dark-energy interpretation [7]. Thus, we focus on this catalogue for comparing different models. By comparing all three configurations (DESY5, Pantheon+, and Union3) we can assess whether any inferred deviation from ΛCDM is driven by the choice of supernova catalogue or represents a robust feature of the combined data.

### B. Numerical tools

As the theoretical platform we use the Boltzmann code `CLASS` [57], which we have modified to implement the modified Cartesian flat model of MTEGR. This includes modifications in both the background equations (13) and (14) together with the modified continuity equation in equation (15) and the linear perturbation equations of Reference [37]. The modified code is publicly available[63]. For the parameter estimation we use the Monte Carlo Markov Chain sampler `Cobaya` [58], which interfaces directly with the modified `CLASS`. The `GetDist` [59] code has been used to analyze the chains and to draw triangle posterior plots.

In the code pipeline, the susceptibility function $S(z)$, and its derivatives, are treated as input functions of the scale factor, and the modified continuity equation (equation (15)) fixes the evolution of all components. The linear perturbation equations of the modified Cartesian flat model were derived and validated against the CMB in Reference [37]; we employ them here without modification, in the synchronous gauge.

The three models we are going to analyze and compare are CPL, the minimal dynamic dark energy model, EXP, short for the exponential ansatz in equation (47), and also the standard ΛCDM model. The CPL and EXP models both have two additional parameters compared to the base ΛCDM model.

## VII. RESULTS

We now report the results of the MCMC analysis of the CMB+BAO+different supernovae catalogs data with the MTEGR model.

Our experiments with multiple functional forms for $\mathfrak{S}(z)$, always select susceptibility functions with the same qualitative shape: $\mathfrak{S}(z)$ starts from 1, develops a dip of depth controlled by $\beta$ in the low-redshift region, and returns to 1 at higher redshifts on a scale set by $\delta$. In most cases the recovered $\mathfrak{S}(z)$ has a minimum at a redshift consistent with the phantom-crossing redshift $z_\times \simeq 0.4$ of the CPL model, and the derivative $d\ln\mathfrak{S}/dz$ changes sign there. Through equation (43), this dip translates into an effective dark energy equation of state that crosses $-1$ from above at $z \simeq z_\times$, precisely the behavior favored by DESI DR2. In other words, the data, when interpreted within the local limit of nonlocal gravity, consistently select a susceptibility with a low-redshift dip, and the dip is the mechanism that generates the phantom crossing.

Fig. 2 presents our main result: a triangle plot of the posterior distributions obtained from the Markov chains generated for the EXP model using the CMB+BAO+DESY5 data combination. The figure shows the credible regions for the cosmological parameters and the EXP model parameters $\beta$ and $\delta$. We find that $\delta$ places the dip in the $\mathfrak{S}(z)$ function near $z_\times \sim 0.4$. On the other hand, the best-fit value of $\beta$, together with its uncertainty, lies more than $2\sigma$ away from the ΛCDM value, $\beta = 0$. A calculation of the AIC and DIC model-selection criteria, presented in Table I, shows that the EXP ansatz is strongly preferred over the baseline ΛCDM model according to both criteria. Compared with the CPL model, however, the EXP ansatz is favored by one criterion and disfavored by the other, indicating that the two models remain competitive.

An important feature of the final posteriors (Fig. 2 and 4) is the correlation between the depth parameter $\beta$ and the Hubble constant $H_0$. This degeneracy follows directly from the background relations of Section IV. Equation (30) reads $H^2(z) = H_0^2\,\mathfrak{S}(z)^{-2}\,[(1+z)^3\Omega_{m0} + \Omega_{\rm de0}]$, so that for a fixed $H_0$ a more negative $\beta$ (a deeper dip in $\mathfrak{S}(z)$) enhances the expansion rate at low redshift, since the factor $\mathfrak{S}(z)^{-2}$ grows above unity where the dip sits. The CMB, however, fixes the angular size of the sound horizon, $\theta_* = r_s/D_A(z_*)$, with $D_A(z_*) \propto (1+z_*)^{-1}\int_0^{z_*} dz/H(z)$. Raising $H(z)$ at low $z$ shortens this comoving distance and therefore enlarges $\theta_*$; restoring the observed value requires lowering $H_0$. The two parameters are thus positively correlated, which explains both the tilt of the $H_0$–$\beta$ contour in Fig. 2 and 4 and the sys-

| Parameter | ΛCDM | CPL | EXP |
|---|---|---|---|
| $\Omega_c h^2$ | $0.118 \pm 0.001$ | $0.119 \pm 0.001$ | $0.118 \pm 0.001$ |
| $\Omega_b h^2$ | $0.0220 \pm 0.0001$ | $0.0220 \pm 0.0001$ | $0.0220 \pm 0.0001$ |
| $H_0$ | $68.424^{+0.305}_{-0.308}$ | $66.760^{+0.591}_{-0.576}$ | $67.096^{+0.473}_{-0.486}$ |
| $\ln(10^{10} A_s)$ | $3.067^{+0.043}_{-0.044}$ | $3.056^{+0.047}_{-0.048}$ | $3.063^{+0.042}_{-0.043}$ |
| $n_s$ | $0.968 \pm 0.003$ | $0.967 \pm 0.004$ | $0.969^{+0.003}_{-0.004}$ |
| $\tau$ | $0.072 \pm 0.022$ | $0.066^{+0.024}_{-0.025}$ | $0.071 \pm 0.022$ |
| $w_0$ | — | $-0.793^{+0.059}_{-0.058}$ | — |
| $w_a$ | — | $-0.594^{+0.233}_{-0.236}$ | — |
| $\beta$ | — | — | $-0.025 \pm 0.007$ |
| $\delta$ | — | — | $0.391 \pm 0.114$ |
| $\Delta\chi^2$ | — | 17.82 | 14.65 |
| ΔAIC | — | −13.82 | −10.65 |
| ΔDIC | — | −13.94 | −16.43 |

TABLE I: Parameter constraints and model comparison statistics for the CMB+BAO+DESY5 data set.

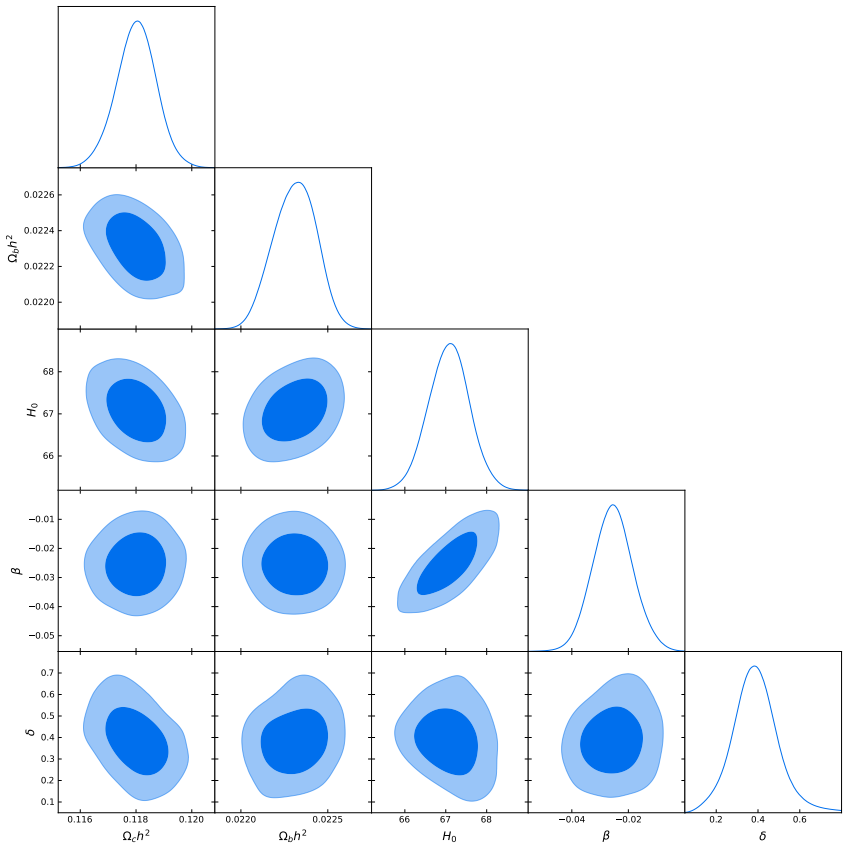


FIG. 2: Posterior distributions for the CMB+BAO+DESY5 data with the exponential ansatz $\mathfrak{S}(z) = 1 + \frac{\beta}{\delta^2} z^2 e^{2(1-\frac{z}{\delta})}$.

tematic shift of $H_0$ among the models in Table I.

The comparison between the results of different models is displayed in Fig. 3, where the constraints on the parameters shared by the three models are overlaid. The figure makes it explicit that ΛCDM, CPL and EXP differ mainly along the $H_0$ direction, while the common cosmological parameters ($\Omega_c h^2$, $\Omega_b h^2$, $n_s$, $\ln(10^{10} A_s)$, $\tau$) remain tightly constrained and essentially model-independent.

To verify that our conclusions do not depend on the particular supernova sample, the EXP MCMC analysis is repeated and reported in Table II for the Pantheon+ [55] and Union3 [56] catalogues, with the CMB and BAO data held fixed. All three catalogues return a negative $\beta$ and a dip position $\delta$ of the same order, resulting in the same qualita-

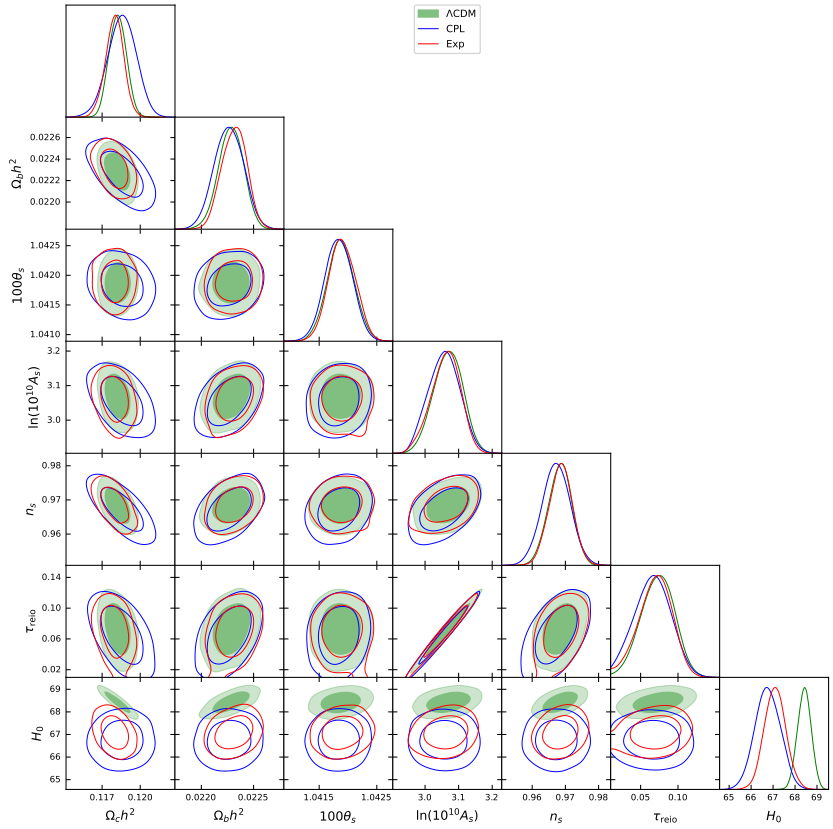


FIG. 3: Complete triangle plot comparing the posterior distributions obtained for the three models ΛCDM, CPL and EXP for the CMB+BAO+DESY5 data set.

| Parameter | DESY5 | Pantheon+ | Union3 |
|---|---|---|---|
| $H_0$ | $67.096^{+0.473}_{-0.486}$ | $67.623^{+0.565}_{-0.551}$ | $66.756^{+0.738}_{-0.716}$ |
| $\beta$ | $-0.025 \pm 0.007$ | $-0.016 \pm 0.008$ | $-0.030 \pm 0.011$ |
| $\delta$ | $0.391 \pm 0.114$ | $0.516^{+0.166}_{-0.158}$ | $0.407^{+0.092}_{-0.098}$ |

TABLE II: Parameter constraints for the exponential model using DESY5, Pantheon+, and Union3 supernovae samples.

tive susceptibility, with only mild shifts in $\beta$ and $\delta$: Union3 prefers the deepest dip ($\beta = -0.030 \pm 0.011$) together with the lowest $H_0$, whereas Pantheon+ prefers the shallowest dip ($\beta = -0.016 \pm 0.008$) and the highest $H_0$. This is precisely the $\beta$–$H_0$ degeneracy described above, and it shows that the low-redshift dip is a robust feature of the data rather than an artefact of a single supernova compilation. The full posterior distributions of $(H_0, \beta, \delta)$ for the three catalogues are displayed in Fig. 4.

## VIII. DISCUSSION

The central result of this paper is that the phantom crossing favored by DESI DR2 can be realized within the local limit of nonlocal gravity without introducing any phantom component. In the modified model, the dark energy fluid has $w = -1$ at all times and satisfies the modified continuity equation (equation (15)); the model does not introduce an explicit phantom matter degree of freedom. What crosses $-1$ is the effective equation of state (equation (43)), a derived quantity that encodes the difference between the true expansion history of the modified model and the standard Friedmann interpretation of that history. This interpretation has a number of appealing features. First, it avoids the well-known instabilities and fine-tuning problems of phantom fields [16]. Second, it makes the crossing generic: any susceptibility with a dip produces a crossing, and the data select a dip.

The physical content of the crossing is the dip in the susceptibility $S(z)$ at $z \simeq 0.4$, which represents a transient, low-redshift feature in the gravitational response of spacetime. This behavior is qualitatively analogous to the frequency-dependent response of a medium in electrodynamics: near a resonance, the real part of a material response function can ex-

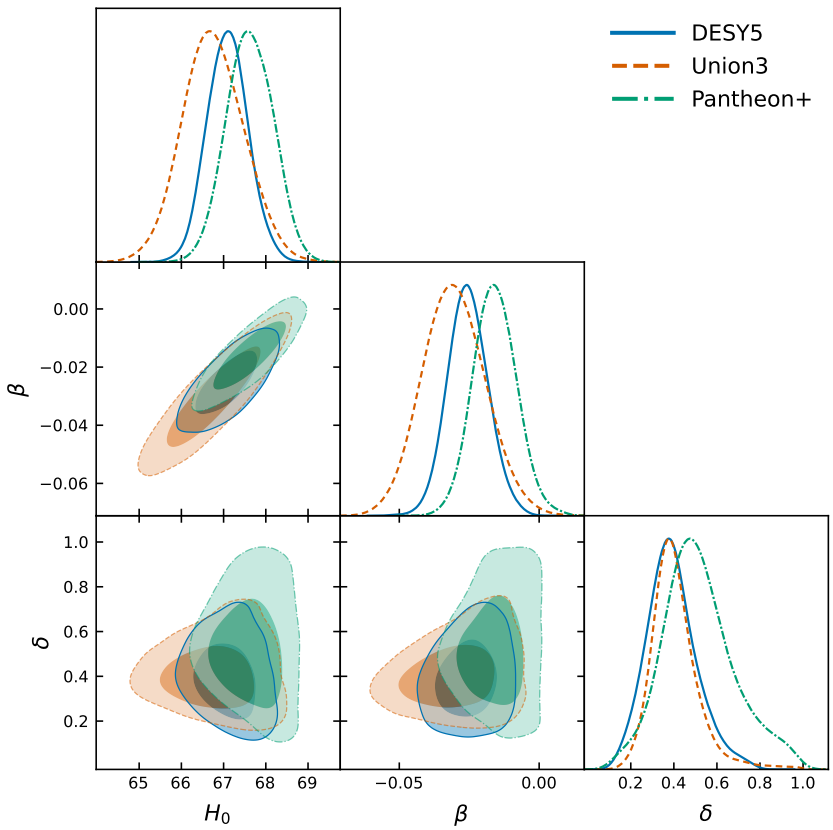


FIG. 4: Complete triangle plot of the joint $H_0$–$\beta$–$\delta$ posterior distributions for the exponential model, comparing the three supernova catalogues DESY5, Union3 and Pantheon+.

hibit rapid variation, including a local extremum or, depending on the model, a change in sign. In a causal medium, the dispersive and absorptive parts of the response are related by the Kramers–Kronig relations [60, 61]. This analogy is intended to be qualitative; $S(z)$ is a gravitational susceptibility and should not be identified directly with an electromagnetic dielectric function.

Several other frameworks have been proposed to explain the DESI DR2 phantom crossing. In scalar-tensor gravity, a reconstruction of the underlying Lagrangian from the DESI DR2 data was performed in Reference [29], where the crossing is realized through a non-minimal coupling and a modified effective gravitational constant; in that model the inferred $H_0$ is also increased relative to ΛCDM. In quintom models, the crossing is achieved with two fields, one quintessence-like and one phantom-like [16]. In string-inspired settings, self-dual potentials have been proposed that mimic the DESI DR2 behavior with a single rolling field near a maximum of its potential [62]. The present framework differs from all of these in an essential way: no new matter component, no second field, and no phantom field is introduced. The crossing is a property of the gravitational sector itself, specifically of the susceptibility $S(z)$, whose role is analogous to that of a constitutive function of a medium. In this respect the model is closer in spirit to the electrodynamics-of-media analogy that motivated nonlocal gravity in the first place [30, 31].

### Data availability

We use public data sets in this work. See Section VI.

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### Appendix A: Absence of a de Sitter solution in MTEGR

Because the impossibility of de Sitter spacetime is the physical origin of dynamic dark energy in this framework, it is worth recalling the argument. Setting $\rho = P = 0$ in equations (13) and (14), the first equation gives, for an expanding universe,

$$\frac{\dot{a}}{a} = \left(\frac{\Lambda}{3}\right)^{1/2}(1+S)^{-1/2}\,, \tag{A1}$$

Differentiating this relation with respect to $t$ and eliminating $\ddot{a}/a$ with the second Friedmann equation, one arrives at

$$2\,\frac{\dot{a}}{a}\,\frac{dS}{dt} = \left(\frac{\Lambda}{3}\right)^{1/2}(1+S)^{-1/2}\,\frac{dS}{dt}\,. \tag{A2}$$

For $dS/dt \neq 0$, division by $dS/dt$ leaves a relation that contradicts the first equation. Hence, for a time-dependent susceptibility, no de Sitter solution exists [33]. We therefore set $\Lambda = 0$ throughout and account for the accelerated expansion through a dynamical dark energy component with negative pressure, $P_{\rm de} = w_{\rm de}\,\rho_{\rm de}$ with $w_{\rm de} < 0$. In the modified Cartesian flat model, $w_{\rm de} = -1$ is allowed (unlike the standard model, where it is trivially the cosmological constant), and equation (17) shows that, even then, the density evolves with $S(t)$.

### Appendix B: Conservation of the effective dark energy

A natural question at this point is whether the effective dark energy defined by equation (36) and equation (39) satisfies a conservation equation. It does not. A direct calculation gives

$$(1+z)\frac{d\bar{\rho}_{\rm de}}{dz} - 3\big(\bar{\rho}_{\rm de} + \bar{P}_{\rm de}\big) = 3\rho_{\rm de0}\,\frac{S(z)-S(0)}{\big[1+S(z)\big]^2}\,. \tag{B1}$$

The right-hand side vanishes only if $S(z) = S(0)$, only if the susceptibility is constant. In general, the effective dark energy is not conserved: the modified continuity equation (equation (15)) redistributes energy among the components in a way that, from the standard viewpoint, looks like an injection or removal of energy into the effective dark energy sector. This is the same phenomenon that, in the modified Cartesian flat model, was identified with the production of heat (or negative entropy production) associated with the variation of $S$ [33].